

D3Net-based All-in-One Restoration for Smartphone Images with Dirty Lenses

SIDL adaptation, paper-baseline comparison, continued training, and failure analysis

DATASET

Problem Setup

- SIDL provides real-world degraded-clean pairs captured with contaminated smartphone lenses.
- Lens contamination causes local artifacts, contrast loss, haze, and frequency-domain distortion.
- This project analyzes D3Net adaptation under a limited class-project training budget.

SIDL validation axis

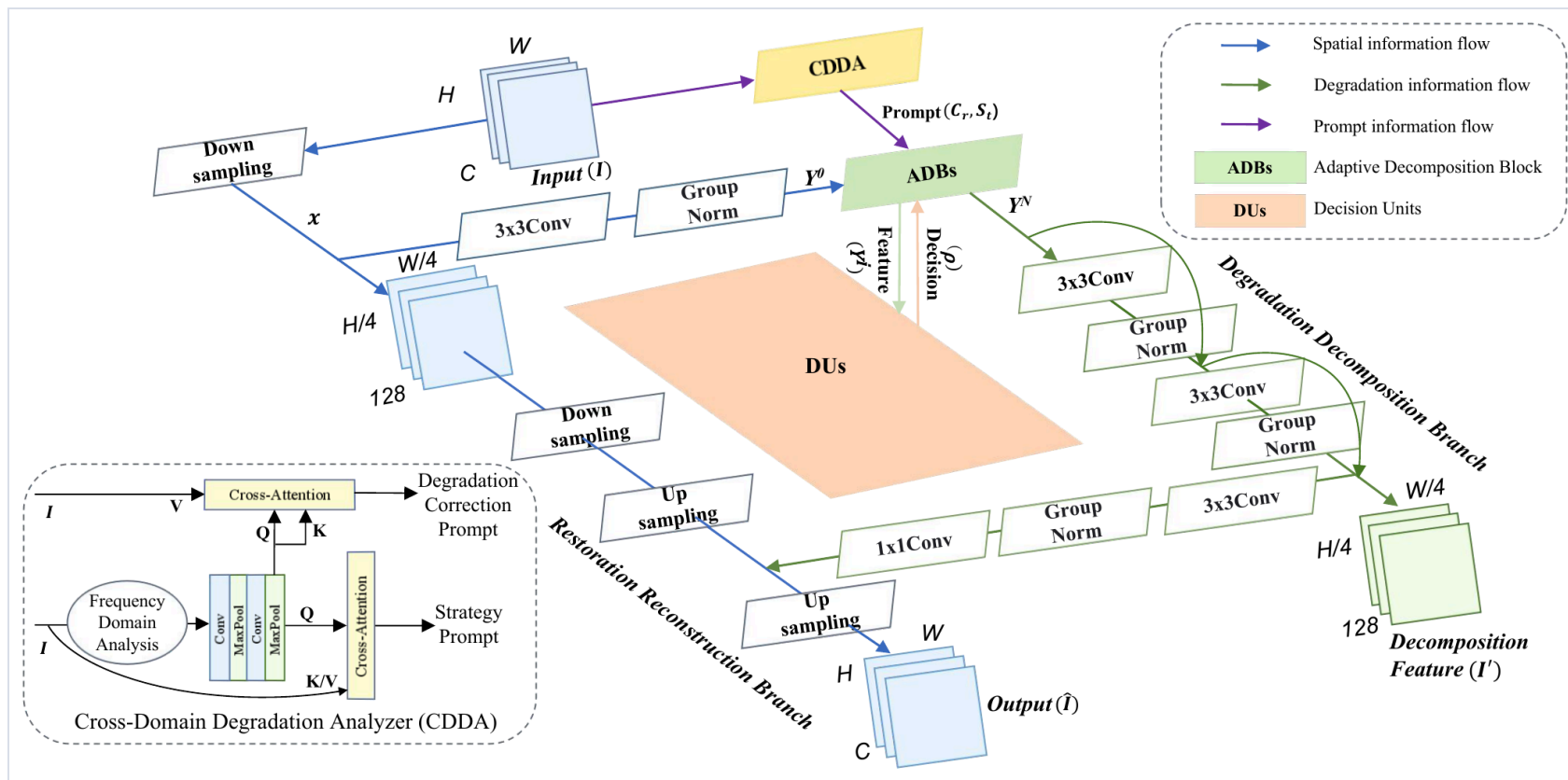
512×512 patchified image pairs

Clean / Dust / Finger / Water / Scratch
/ Mixed

Easy / Medium / Hard difficulty

Metrics: PSNR / SSIM

D3Net Architecture



Core components

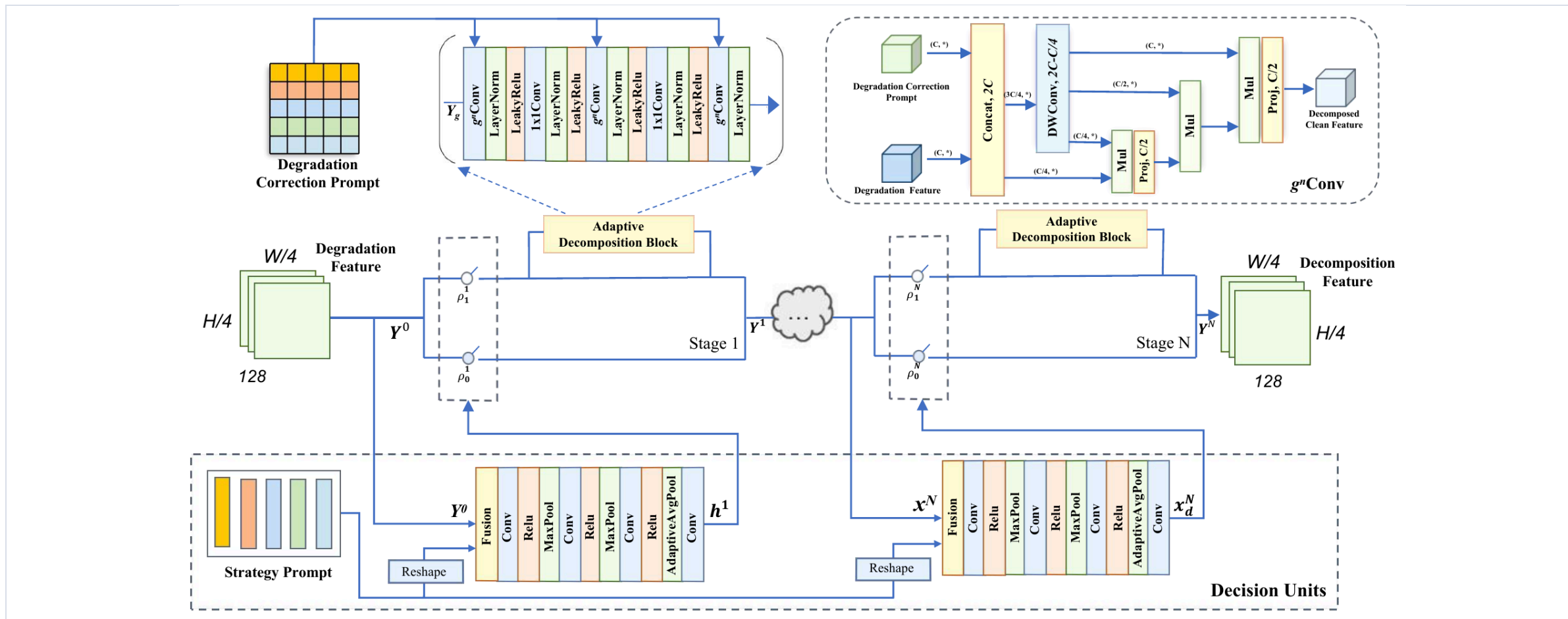
Residual prediction: restored image = input + residual

Fourier-domain cues for degradation representation

Dynamic degradation decomposition for mixed contamination

The model is used as an all-in-one backbone rather than a degradation-specific model.

Dynamic Decomposition



D3Net uses prompt-conditioned feature processing so that local and mixed contamination can be handled within one restoration model.

Experimental Protocol

Main setting

256×256 random crop, scratch training, batch 8

Validation

512×512 center crop, image-wise PSNR / SSIM

Optimization

L1 loss, Adam, cosine LR, single RTX 4090

D3Net original

Official setting is roughly 1200 epochs

Our best

Continued 256 training reached epoch 73 best

Evaluation

Residual output, denormalize, clamp to [0,1], then compute PSNR/SSIM

BASELINE COMPARISON

Baseline Comparison

METHOD	LEVEL	CLEAN	DUST	FINGER	WATER	SCRATCH	MIXED	AVG.
AirNet	Easy	27.10	23.40	25.10	24.85	25.84	25.30	25.26
	Medium	26.13	19.69	22.53	20.36	21.24	16.79	21.12
	Hard	23.79	16.18	15.83	16.64	18.13	14.48	17.50
DiffUIR	Easy	34.51	26.96	28.82	27.42	30.14	28.22	29.34
	Medium	33.25	22.11	26.16	24.65	27.07	20.32	25.36
	Hard	29.47	20.38	18.93	19.91	21.27	18.97	21.71
D3Net ours	Easy	32.78	25.07	24.59	26.38	27.84	23.32	26.66
	Medium	30.26	24.03	23.69	24.05	27.41	22.56	25.33
	Hard	27.70	17.05	20.23	20.22	20.38	17.98	20.59

AirNet/DiffUIR are SIDL paper test-set values. D3Net is our epoch-73 validation result, so this is task context rather than a strict leaderboard.

RESULTS

Ablation Study

RUN	INIT.	CROP	EPOCH	PSNR	SSIM
128 baseline	scratch	128	34	23.33	0.8357
256 main	scratch	256	34	23.82	0.8398
256 continued	latest	256	73	23.86	0.8419
256 low-LR	256 best	256	3	23.76	0.8409
512 refinement	256 best	512	2	23.27	0.8336

12823.33

25623.82

256 cont.23.86

256 polish23.76

51223.27

INTERPRETATION

Cross-Difficulty Trade-off

Overall score improved from 23.82 / 0.8398 to 23.86 / 0.8419.

Hard average improved from 20.33 to 20.59 dB.

Easy PSNR decreased from 27.00 to 26.66 dB, suggesting mild over-correction.

Key limitation

The original D3Net training scale is roughly 1200 epochs. Our best checkpoint is epoch 73, so the result should be interpreted as limited-budget adaptation.

The training loss is L1 on normalized tensors, while evaluation uses PSNR/SSIM on denormalized [0,1] images. This loss-metric mismatch may affect convergence.

Failure Analysis

TYPE	AVG. PSNR	AVG. SSIM
Clean	29.87	0.9125
Dust	21.90	0.8217
Finger	21.97	0.7897
Water	22.87	0.8433
Scratch	25.08	0.8505
Mixed	20.67	0.8311

Hard Dust **17.05 / 0.7086**

Hard Mixed **17.98 / 0.7900**

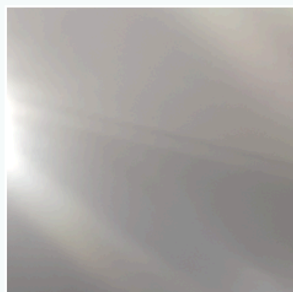
Severe local scattering and ambiguous mixed contamination remain difficult to separate from true image structure.

QUALITATIVE RESULTS

Qualitative Results

1. finger / hard / Case051_F_001.png

PSNR 19.28 dB SSIM 0.9316



Input



Output



GT

2. dust / medium / Case251_D_018.png

PSNR 19.25 dB SSIM 0.7943



Input



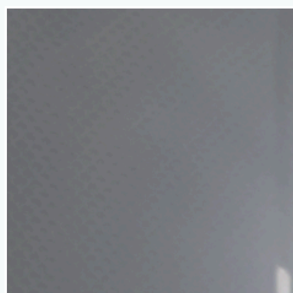
Output



GT

3. scratch / hard / Case073_S_001.png

PSNR 10.47 dB SSIM 0.6080



Input



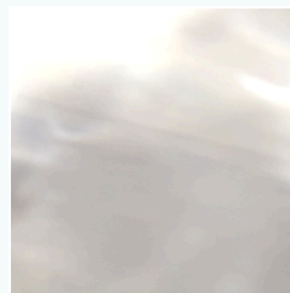
Output



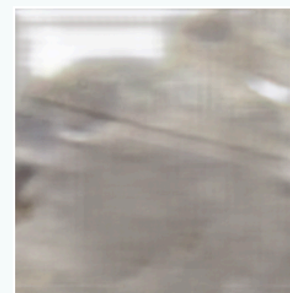
GT

4. water / hard / Case051_W_001.png

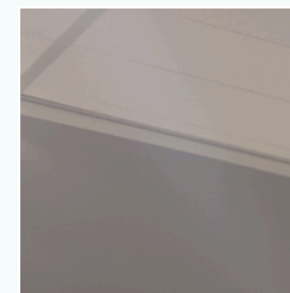
PSNR 17.70 dB SSIM 0.8999



Input



Output



GT

Each panel shows input / D3Net output / GT using original report assets.

Summary & Future Work

Main result

Epoch 73 continued 256 training: 23.86 dB / 0.8419

Key finding

Hard restoration improved, while Easy PSNR slightly decreased.

Limitation

The epoch budget is far smaller than the original D3Net training scale.

Future work

Long training, pretrained D3Net, 512 curriculum, Charbonnier/L1+MSE losses, and TTA